

Commentary

Critic and Conscience of Society:

A Reply to John Hattie

Ivan Snook, John Clark, Richard Harker, Anne-Marie O'Neill and John O'Neill
Massey University.

Following the release of John Hattie's book, *Visible Learning* (2009), we published an article in this journal (Snook et al, 2009). In a recent issue of NZJES, Hattie (2010) wrote a spirited reply to those who in various ways have criticised his research, his public statements, his political activities and his personal motivation. He located this reply within a discussion of the role of academics as "critics and conscience of society." Given that he has been subjected to such varied attacks, it is quite reasonable that he should do so but, as a result of his approach, our "sympathetic critique" was not clearly distinguished from other more personal and sometimes overtly political attacks. It is not for us to defend all those who, in various ways, have criticised his research and his involvement in policy. We are concerned simply with those sections of his reply which refer to our article, directly or indirectly.

Our agenda was set out at the beginning of our article:

The announcement of the book has already led to a good deal of discussion both in New Zealand and overseas and seems to have captured the attention of policy makers. It is, therefore, important that members of the educational research community pay John Hattie the courtesy of subjecting his conclusions to critical scrutiny in a spirit of mutual truth seeking to ensure that:

1. Discussions are based on a careful reading of the book, rather than on half baked "reactions" in the popular media;
2. The caveats which Hattie himself sets out are carefully noted so that decisions are not made in opposition to the message of this book and
3. The findings are not "appropriated" by political and ideological interests and used in ways which the data do not substantiate. (p. 93)

We believe that a careful reading of our article will show that we restricted ourselves to this agenda: We focused on the findings in his book and on their potential use for political purposes. We agree with Hattie that there is a pressing need in New Zealand for university academics to openly debate the differences between intellectuals (Said, 1994) and policy entrepreneurs (Mintrom, 1997). Indeed, in a small democracy like New Zealand where funding for educational research is almost exclusively focused on Ministry policy implementation or evaluation (OECD, 2001),

it is a moot point whether it is possible for public academics to be both inside and outside the system (Hattie, 2010, p. 93). However, that debate is for another occasion as are other important debates arising from the many issues raised by Hattie in his wide-ranging commentary.

We made no “personal attacks” and should not be confused with those who “uncritically respond through new media such as blogs” (Hattie, 2010, p. 85) or with those who “had not read the book” (Hattie 2010, p.87). We read the book carefully and, we believe, presented a balanced commentary. He suggests that, “critique of ideas...is the essence of academia and can lead to debate which is more informed.” (Hattie, 2010, p. 87). We agree with this and hoped to initiate debate about his ideas in the pages of an academic journal. We believe that our account focused on the ideas rather than the person and that we presented our critique with the “courtesy and goodwill” which he believes (and we believe) should characterise academic debate. In this reply, therefore, we focus our attention on those criticisms which seem to involve us.

Hattie says: “Among the most common of criticisms of *Visible Learning* is the place of non-school factors” (Hattie, 2010, p 89). His response to such criticisms is that he is well aware of non-school factors but preferred to ignore them and to concentrate on what can be done in schools. We acknowledged this to be his position and indeed quoted the very line which he quotes back at us: We pointed out that he downplayed non-school factors “NOT because they are unimportant, indeed they may be more important than many of the issues studied in this book. It is just that I have not included these topics in my orbit” (Hattie, 2009, pp. x-xi; Snook et al, 2009, pp. 94-95; Hattie 2010, p. 89), but we went on to say, “It is easy for those seeking to make policy decisions to forget this significant qualification” (Snook et al, 2009. p. 98). We then cited research which indicates that home background is indeed the most important variable affecting school achievement, and indeed is more important than any of the variables discussed in his book. We pointed out that Hattie’s own work testifies to this.

He argues that his comment about not writing a book about “criticism of research . . . seems to have been misinterpreted as my paying no attention to the quality of the research” (Hattie, 2010, p. 88). He says that where research quality relating to specific influences has been shown to make a difference this has been cited and where meta-analyses have been poor this has been pointed out. However:

1. In our original article, we carefully noted that there are occasions on which he comments on the quality of research, most notably in relation to “learning styles” where we say that he is “justifiably suspicious of the motives behind much of research and appropriately sceptical about the results” (Snook et al, p.95). We also state, “His own approach acknowledged that a simple effect size analysis cannot be maintained, when there are questions about the underlying quality of some of the studies he includes” (Snook et al, p. 95).
2. Nevertheless his original disclaimer is actually much stronger than he now suggests: “I have deliberately not included much about moderators of research findings based on research attributes (quality of study, nature of design) again not because they are unimportant . . . but because they have been dealt with elsewhere by others” (Hattie, 2009, p. ix). We submit that we

were right to state that, despite some exceptions, “he does not subject the various studies to critical analysis” (Snook et al, 2009, p. 95) and we note that he himself raised this as a problem with his analysis.

He goes on to say that: “Snook et al (2009) claimed that I did not acknowledge that ‘bias is not normally controlled in meta-analysis’ but they failed to acknowledge the rich literature on this topic which shows how controlling bias is a major concern of this method” (Hattie, 2010, p. 88). But the only way in which Hattie defends his work against this charge is by repeating our own point that “the studies are heavily weighted to published studies”. He argues that peer review of material for publication is an important way of avoiding bias. But has been often pointed out, relying only on published work can also *lead to* bias in that studies which do not show marked effects or effects in the “wrong direction” frequently do not make it into publication. The possibility of bias is not negated by talk of “the rich literature” on the topic.

He suggests that we claimed that he ignored the many potential influences (such as quality, year of schooling) on his overall conclusions (Hattie, 2010, p. 88). We did not claim that he ignored these variables. On the contrary, we acknowledged (e.g., in relation to homework) that Hattie presents the data with all the relevant nuances (age of student, ability of student, subject studied, etc.). However, in the end, he produces an average effect size (in this case .29) which means that a signal is sent out to the profession (and certainly to media commentators) that “homework is not important”, even though his own data reveal that homework is very significant for students in secondary schools, for high ability students and in some subject areas. Thus, it is obvious that abolishing homework might impair the education of many students. This reinforces our belief that caution is needed before deducing policies from meta-analyses.

John Hattie argues, “Snook et al claimed that New Zealand may be so different from the rest of the Western world that we ‘should exercise great care in relating the meta-analyses to New Zealand education’ ” (Hattie, 2010, p.89). Once again, it was Hattie himself who drew attention to the problem of generalising from largely USA sources to other education systems for example those in developing countries. We simply noted, “It could easily be that New Zealand schools, teacher, students, and curricula are sufficiently different from those of the USA that the meta-analyses may not fully apply in this country” (Snook et al, 2009. pp. 97-98). This is a very modest and reasonable claim.

He further suggested that we accused him of adopting an “arbitrary” cut off point at 0.40. All cut off points can be said to be in some way arbitrary (as he admits); we were simply suggesting that for many purposes a lower cut off can be justified (as he acknowledged in relation to drug research). We went on to make the more important point that:

Much depends on the quality of the research studies in the various meta-analyses. If the sample is large and random...a “small” effect size is of considerable significance. On the other hand, large effect sizes from small samples are meaningless at best and positively dangerous when joined to other studies to produce an “average”. (p. 94)

We do acknowledge, however, that our point about the .17 effect size from Outward Bound programmes was based on a misunderstanding, which we regret.

Hattie says: "Snook et al claim that I am more concerned with surface level achievement" (Hattie, 2010, p.90). We did not say (or imply) that Hattie is more concerned with surface level achievement rather than "conceptual knowledge". On the contrary, we noted that "conceptual knowing or understanding is what [Hattie] thinks should be the result of good teaching" (Snook et al, 2009, p. 95). We also repeated his own reservation that so much of the *research* is based on surface level achievement and concluded that "there is less to be drawn from his synthesis than commentators have suggested". This is a warning to policy makers rather than a criticism of John Hattie's work.

In relation to class size, to which we devoted quite a lot of space, he accuses us of being "selective" in choosing our studies and again defends the merits of focusing on averages: "The major advantage of meta-analysis is that extreme findings are moderated by a larger sample of studies" (Hattie, 2010, p. 90). Here, it seems that we have a real difference between us. We recognise the usefulness of meta-analyses but are extremely sceptical of too much reliance on them in making practical judgments. The two cases we cited as showing favourable results from lowered class sizes were large-scale, well designed (one involved a random design), and carried out over several years. We are not sure where Hattie would stand, but we put more faith in such studies than in "averages" from multiple studies, many of which may be of poor quality. In the Murnane and Levy (1996) study, the authors make the point that if the results from all schools had been averaged there would have been "a small negative result" but the two schools which make other changes in addition to class size experienced striking positive results. Such points may indeed be "well rehearsed" in the literature on meta-analyses but, to our minds, they are compelling. It is, however, worth noting that Hattie and we are in total agreement on one matter: Smaller classes *allow* but do not *ensure* more effective teaching. It would certainly be quite wrong to conclude that class size is irrelevant to student achievement. Once again, there is an important point here about using meta-analyses for deriving policy.

We are rather surprised that Hattie did not comment on what is perhaps *the* major problem in using meta-analysis in educational research. The technique had its origin in medical science with its carefully defined concepts and tight (usually experimental) research design. In education, however, the variables being studied are often poorly conceptualised and the studies often far from rigorous. How does one clearly distinguish for research purposes between a classroom that is "teacher centred" and one which is "student centred" and, in comparing them, how can one control all the variables in a noisy and busy classroom with perhaps more than 30 participants? (Snook et al, p. 97). It is interesting that after forty years of close-grained research in classrooms, Graham Nuthall (2007) concluded that "although there are many published reports of educational research, you need to be very selective in identifying those that have something trustworthy to say about classroom teaching and learning" (p.16). In this connection, it is interesting to juxtapose the title of John Hattie's book, *Visible Learning*, based on international meta-analyses with that of Graham Nuthall, *The Hidden Lives of Learners*, based on decades of research in New Zealand classrooms.

Conclusion

In our view, John Hattie's article has not satisfactorily addressed the concerns we raised about the use of meta-analyses to guide educational policy and practice. If there is to be sensible public debate in New Zealand on such matters, the utility of the underlying ideas must be debated in a genuine spirit of scholarly truth-seeking. To conclude this reply, we summarise our concerns both with Hattie's 2009 synthesis, and his subsequent response to the issues we raised.

The methodology involved in meta-analyses often obscures rather than illuminates what is needed for good policy decisions. The grouping of various factors into one "effect size" often hides the complexity of the issues. We tried to illustrate this by reference to the findings on homework and class size.

The results reported in meta-analyses of educational research tend to mask complex interactions between schools, students, families and communities. Without this awareness, lists of "effect sizes" merely encourage silver bullet responses to complex educational problems.

Many case studies of successful schools and classrooms show *actual* interaction effects at work. One insightful intervention can counter other influences even when these have greater effect sizes. Meta-analyses can be seriously misleading unless the interaction effects are taken into account.

For these reasons we fully subscribe to the position of John Dewey, which Hattie himself quotes with approval in his book: "Evidence does not supply us with rules for action but only with hypotheses for intelligent problem solving, and for making inquiries about our ends in education" (Hattie, 2009, p.247). Meta-analyses and other forms of educational research summaries can rarely be directly applied to educational policy or practice. We urge caution lest teachers and teacher educators use the findings in a simplistic way and politicians find facile (but misguided) solutions to complex educational problems.

In relation to teacher education, we can only repeat what we wrote in our original article:

It is important that students preparing for teaching learn about the research process and how easily it leads to error rather than truth. They need to respect research but be acutely aware of its limitations. The research that they need to know about goes beyond what happens in schools and classrooms. As this review has shown, what students bring from their social class, family, culture, home background and prior experiences is more important than what happens in the school, even though what happens in the school (particularly what teachers are and do) is very important. The secret of school improvement lies in the recognition of these factors and their integration into a social, economic and educational programme. (p. 105)

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